World Journal of Surgical Oncology

RESEARCH

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Real-time deep learning for tumor segmentation and tool tracking: development and validation of an AI navigation system in vacuum-assisted breast biopsy

Xinran Shao[1†], Yunzhi Shen[1,2†], Pingdong Sun[1], Yihan Sun[3], Xingai Ju[4], Hongjie Zhu[5], Hong Li[5], Qiushi Li[1], Ruan Ting[6], Jinrui Liu[1], Yuqing Wang[1], Qikun Guo[7], Yuxin Ma[7], Xiang Fei[1*], Hang Sun[8*] and Jianchun Cui[1*]

Abstract

Background Vacuum-assisted breast biopsy (VABB) is a widely adopted minimally invasive technique for the diagnosis and treatment of breast lesions. However, the procedure heavily relies on real-time ultrasound guidance, posing significant challenges for junior surgeon who lack radiological experience in accurately localizing both the lesion and the biopsy needle. Currently, there is no dedicated real-time artificital intelligence (AI) navigation system specifically designed for VABB procedures.

Methods We developed a novel two-stage real-time AI navigation system based on the YOLOv11 deep learning architecture. Model 1 performs initial localization of the tumor and the cutter slot, while Model 2 focuses on real-time tracking of the knife and tumor during resection. The system was trained and validated using 22,278 annotated ultrasound images from 167 VABB procedures conducted at People's Hospital of China Medical University. A rigorous three-fold cross-validation was implemented to assess model performance, and the localization accuracy was compared with that of junior surgeon. Additionally, we evaluated the system's real-time processing performance on both GPU and CPU platforms.

Results The AI system demonstrated superior performance across all evaluated metrics. ForModel 1, the mean Average Precision at IoU threshold 0.5 (mAP50) for tumor detection and groove localization reached 0.907 and 0.671, respectively, significantly outperforming junior surgeon (0.551 and 0.120). For Model 2, the mAP50 for tumor

†Xinran Shao and Yunzhi Shen are contributed equally to this work. as co-first authors.

*Correspondence:
Xiang Fei
fx057619@sina.com
Hang Sun
sunhang@sylu.edu.cn
Jianchun Cui
cjc7162003@aliyun.com

Full list of author information is available at the end of the article

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and needle tip tracking were 0.829 and 0.765, respectively, compared to 0.758 and 0.350 achieved by surgeons. The system achieved a real-time processing speed of 1.2 ms per frame on GPU and 32.6 ms per frame on CPU.

Conclusion This study presents the first dedicated AI-based navigation system for VABB, showing substantial improvement in localization accuracy over manual operation by junior surgeon. The system's robust detection capability and real-time performance highlight its strong potential for clinical application, especially in surgical training and complex cases requiring precise instrument control.

Trial registration Clinical Trial Registration No. 2022JH2/101300026

Keywords Artificial intelligence, Vacuum-assisted breast biopsy, Computer-assisted surgery, Deep learning, Surgery navigation systems, Breast tumor

Background

Vacuum-assisted breast biopsy (VABB) has a 30-year history since its introduction in 1994 [1, 2]. Based on the principle of negative-pressure suction and guided by ultrasound imaging, this technique utilizes a high-speed rotating cutting needle to precisely excise breast lesions [3, 4]. At present, VABB has been widely applied in the excision of benign breast tumors, pathological sampling of suspicious malignant lesions, and the management of residual disease following neoadjuvant therapy for breast cancer [5, 6]. Compared to open surgery, VABB offers several advantages, including minimal invasiveness, rapid recovery, improved cosmetic outcomes, and a lower incidence of postoperative complications [7, 8]. However, the widespread adoption of this technique faces multiple challenges, particularly the lack of proficiency in breast ultrasound imaging among less experienced surgeons [9, 10].

The emergence of artificial intelligence (AI)-assisted intraoperative navigation technologies provides promising solutions to these challenges. AI techniques enable more accurate medical image segmentation [11, 12], while intraoperative navigation systems allow for real-time acquisition and analysis of anatomical data, offering precise surgical guidance and localization support to surgeons [13, 14]. These technologies not only enhance surgical accuracy and safety but also reduce operative complexity, shorten procedure time, and lower postoperative complication rates [15, 16]. Moreover, AI navigation systems hold potential in supporting junior surgeons in acquiring key surgical skills more rapidly, thereby shortening the learning curve [17–19]. Recently, such technologies have been introduced in various ultrasound-guided breast surgeries, including ultrasound-guided breast-conserving surgery for breast cancer [20] and ultrasound-guided breast needle biopsy [21]. However, there is currently a lack of AI-assisted navigation systems specifically tailored for VABB procedures.

To address this gap, the present study developed a real-time intraoperative navigation model based on the YOLOv11 deep learning framework, designed to simultaneously localize both the rotating cutting needle and breast lesion by dynamically analyzing ultrasound imaging data. Furthermore, the effectiveness of this AI-assisted decision-making system was validated by comparing its performance with that of junior surgeon in identifying the cutting needle and lesion during the procedure.

Method

Study design

This study strictly adhered to the ethical principles outlined in the Declaration of Helsinki. The study protocol was reviewed and approved by the institutional ethics committee (No. 2023H005; Clinical Trial Registration No. 2022JH2/101300026). Informed consent was obtained from all patients enrolled in the study. The research initially involved the processing of ultrasound video images to establish a structured dataset. A YOLOv11-based deep learning model was developed to construct a real-time intelligent guidance system for accurate intraoperative localization of the vacuum-assisted biopsy groove and tumor. The system consisted of two modes: Model 1 was designed to assist in determining the relative positions of the tumor and cutter slot (localization phase), while Model 2 supported the resection process following localization. A comparative experiment was ultimately conducted to evaluate the identification capabilities of the model versus junior surgeon in recognizing the knife and tumor.

Dataset

This was a single-center retrospective study. Ultrasound video data from 167 patients undergoing VABB at the Department of Breast and Thyroid Surgery, People's Hospital of China Medical University, were collected between January 2022 and September 2024. The imaging systems included GE-KOGIQP3 and Mindray E1Exp ultrasound machines. All procedures were performed using the Bard Encor DR-ENCOR system (USA), with a 7-gauge ECP017 VABB knife. Inclusion criteria were: (1) tumors classified as BI-RADS 4a; (2) BI-RADS 3 tumors with surgical indications; and (3) patients with single or multiple tumors in which only one required excision. Exclusion criteria

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included: (1) incomplete clinical data; and (2) poor-quality surgical video recordings.

Preprocessing

A systematic video preprocessing workflow was implemented. Surgeons reviewed full-length surgical videos and manually extracted segments from the moment the knife was inserted until the tumor was positioned for optimal resection. Key frames were extracted at 10-s intervals, generating a dataset of 11,610 images for constructing Model 1. Dataset 2 was composed of 10,668 images obtained from segments focusing on the resection phase, with one key frame extracted every 10 frames. Both datasets were split according to a 7:2:1 ratio into training (116 cases), validation (33 cases), and test sets (18 cases). Image annotation was conducted using the X-Anylabeling software (CPU version V2.5.0), developed by CVHub520 Laboratory. In Dataset 1, bounding boxes were used to annotate the location of the tumor and cutter slot. In Dataset 2, bounding boxes labeled the tumor and knife. All surgical video segmentation and key frame annotation were completed by two breast surgeons with more than 5 years of VABB experience. Discrepant or difficult cases were adjudicated by a third surgeon with over 10 years of VABB experience.

Model

The YOLO architecture has evolved from its early speed-focused versions (YOLOv1–v3) to more advanced frameworks (YOLOv4–YOLOv11), integrating attention mechanisms, multi-scale detection, and feature fusion to enhance object localization and classification accuracy [22, 23]. This study employed YOLOv11, the latest in the YOLO series, to develop an AI-based surgical navigation model featuring enhanced performance in object detection, segmentation, and pose estimation tasks due to its architectural innovations [24, 25].

Comparison between model and junior surgeon

During model validation, key frames from the independent validation sets of both models were selected for human–machine comparative analysis. For Model 1, a junior surgeon (less than 5 years of VABB experience) independently annotated the tumor and cutter slot in the validation set. These annotations were compared to reference annotations generated by two senior surgeons (more than 5 years of VABB experience) to evaluate differences in localization accuracy. For Model 2, the same junior surgeon annotated the tumor and knife during the dynamic resection phase. These results were compared to the expert reference standard to assess the coordination accuracy of tumor and knife recognition between the model and human readers.

Hardware specifications

All computations were performed using the PyTorch framework, with reproducibility ensured by setting a fixed random seed. Model training scripts were implemented in Python 3.6 (Python Software Foundation, Wilmington, DE). The hardware setup included an Intel® Core™ i9-14900KF processor and an NVIDIA® GeForce RTX™ 4090 GPU with 24 GB of memory.

Evaluation metrics

Model performance was primarily evaluated using the mean Average Precision at a 0.5 Intersection-over-Union threshold (mAP50), precision, and recall. Additionally, the F1-score, Dice Similarity Coefficient (DSC) and Hausdorff Distance (HD) were reported to provide a complementary assessment of classification balance and boundary delineation accuracy. For each sample, we performed a one-to-one matching between the "gold standard'" and the "surgeon/model annotations", considering detection accuracy, classification correctness, and localization precision. The results were represented by AccCoef. These metrics were recorded during each fold of three-fold cross-validation. To facilitate intuitive performance interpretation, precision-recall curves and confusion matrices were plotted. In addition, inference speed (average processing time per frame) and memory consumption were assessed on both Intel® Core™ i9-14900KF and NVIDIA® GeForce RTX™ 4090 platforms.

Statistical analysis

Statistical analyses were performed using SPSS Statistics software (version 26.0, IBM Corp., Armonk, NY, USA). To evaluate the difference in performance between the surgeon and model, a paired t-test was conducted. A 95% confidence interval (CI) for the mean difference in AccCoef between the surgeon and model was calculated. A p-value of less than 0.05 was considered statistically significant for all tests.

Results

Baseline

The characteristics of the 167 patients are summarized in Table 1.

A total of 167 patients were included in the analysis. The median age was 44 years (interquartile range [IQR], 34.5–50), and the median lesion size was 10 mm (IQR, 8–15). According to BI-RADS classification, 66 patients (39.52%) were categorized as category 3, and 101 patients (60.48%) as category 4a.

Model 1

The recognition performance of the model for tumors and the cutter slot is illustrated in Fig. 1. A quantitative comparison was conducted between Model 1 and the

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Table 1 Patient characteristics in training and testing datasets

Characteristic	Dataset
Age (years)	44(34.5–50) [a]
< 40	67
40–49	56
49 >	44
Tumor (mm)	10(8–15) [a]
< 10	69
10—19	76
> 19	22
Operation time (min)	10(6–18)[a]
BI-RADS	
3	66(39.52%)
4a	101(60.48%)

[a]Median (interquartile range)

junior physician (Table 2). For tumor detection, Model 1 achieved a precision of 0.886, markedly higher than the junior surgeon's 0.582. In terms of localization accuracy, the model reached a mAP50 of 0.907, substantially outperforming the physician's 0.551. For cutter slot identification, Model 1 also showed superior performance, with a precision of 0.759, recall of 0.638, and mAP50 of 0.671, compared to the physician's 0.192, 0.435, and 0.120, respectively. When evaluating overall performance across all categories (tumor and cutter slot), the model achieved a precision of 0.822, recall of 0.747, and mAP50 of 0.789, whereas the junior surgeon obtained only 0.387, 0.653, and 0.335, respectively. Model 1 demonstrated significantly higher accuracy, with a mean AccCoef of 0.816, compared to 0.609 for the doctor. Statistical analysis revealed a highly significant difference between the two methods (paired t-test, $p < 0.001$), with the Model 1 outperforming the doctor by an average of 0.20 to 0.22 (95% CI: 0.193–0.221). These results highlight the model's significant advantage in tumor localization accuracy and cutter slot detection reliability. Model 1 demonstrated superior performance over the junior surgeon baseline, with notably higher F1-score (0.765 vs. 0.486), improved DSC (0.714 vs. 0.525), and substantially reduced boundary error reflected by a lower HD (205.65 px vs. 376.86 px).

We further performed a three-fold cross-validation to objectively assess the model's robustness, with results summarized in Table 3. The average mAP50 (all) for Model 1 reached 0.789. The confusion matrix in Fig. 2 illustrates class-wise classification outcomes, with true positive rates of 0.72 for tumor and 0.70 for cutter slot. Figure 3 presents the precision–recall (P–R) curves, which evaluate the balance between precision and recall at various thresholds. Notably, the model achieved a high precision of 0.823 for tumor classification.

Model 2

The recognition performance of Model 2 for tumors and the knife is illustrated in Fig. 4. A quantitative comparison between Model 2 and the junior physician was conducted (Table 4). Model 2 achieved a precision of 0.804 for tumor detection, slightly outperforming the junior surgeon's 0.785. In terms of localization accuracy (mAP50), the model reached 0.829, which exceeded the physician's 0.758. For knife recognition, Model 2 demonstrated a clear advantage, with a precision of 0.830, recall of 0.749, and mAP50 of 0.765, compared to the surgeon's 0.502, 0.606, and 0.350, respectively. Across all categories, the model maintained superior overall performance, with a precision of 0.817, recall of 0.751, and mAP50 of 0.797, while the junior surgeon achieved 0.644, 0.735, and 0.554, respectively. Model 2 achieved a mean AccCoef of 0.767, while the junior surgeon achieved only 0.599. Statistical analysis showed a highly significant difference in performance (paired t-test, $p < 0.001$), with Model 2 outperforming the junior surgeon by an average of 0.16 to 0.18 (95% CI: 0.157 – 0.178). These results indicate that Model 2 outperformed the junior surgeon in key metrics such as tumor localization accuracy and knife tip recognition consistency. Model 2 exhibited a marked improvement over the junior surgeon baseline, evidenced by a significantly higher F1-score (0.791 vs. 0.512), a substantial gain in DSC (0.751 vs. 0.507), and a dramatic reduction in

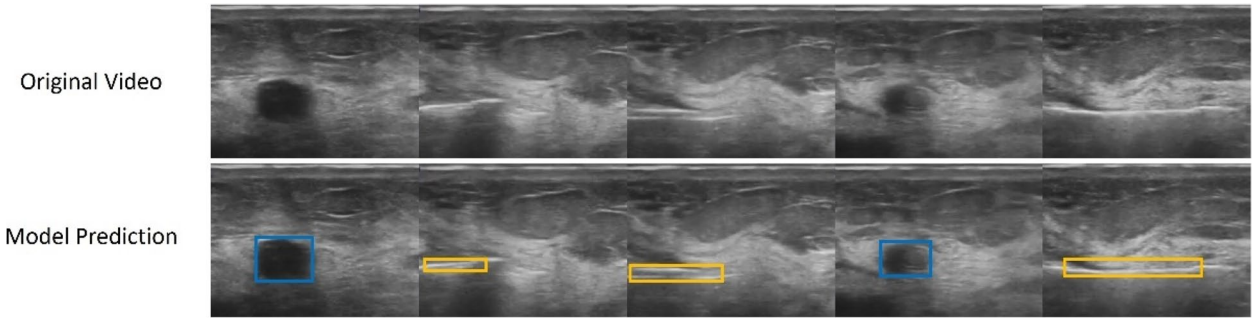


Fig. 1 Schematic Diagram of Model 1 Recognition (within the same case video stream). Yellow boxes indicate the cutter slots identified by the model in ultrasound videos, while blue boxes denote tumors detected by the model

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Table 2 Performance of Model 1 and comparison with junior surgeon

	Precision(tumor)	Recall(tumor)	map50(tumor)	Precision(knife)	Recall(knife)	map50(knife)	Precision(all)	Recall(all)	map50(all)	AccCoef(all)
Yolo11	0.886	0.856	0.907	0.759	0.638	0.671	0.822	0.747	0.789	0.816
Junior Surgeon	0.582	0.872	0.551	0.192	0.435	0.120	0.387	0.653	0.335	0.609

boundary error with a far lower HD (195.20 px vs. 309.81 px).

The results of three-fold cross-validation, which reflect the model's generalizability, are summarized in Table 5. The average mAP50 (all) for Model 2 reached 0.801. The confusion matrix in Fig. 5 shows classification results for each category, with true positive rates of 0.74 for tumor and 0.81 for knife. The P–R curve in Fig. 6 illustrates the balance between precision and recall across thresholds. The overall mAP50 of 0.801 under an IoU threshold of 0.5 further confirms the model's strong performance.

Real-time performance

The YOLOv11-based navigation model demonstrated efficient computational performance, with a total of 9,413,574 parameters and a computational complexity of 21.3 GFLOPS. In measured inference speed tests, the model required 32.6 ms per frame on the CPU platform and only 1.2 ms per frame on the GPU platform, meeting the deployment requirements for practical clinical applications.

Discussion

This study proposes the first real-time navigation system for VABB based on the YOLOv11 deep learning framework. The system includes Model 1, which addresses the localization phase during ultrasound-guided VABB, and Model 2, which facilitates the tumor resection process. Comparative analysis against junior surgeon demonstrated that both models outperformed less experienced clinicians in intraoperative recognition of tumors and the biopsy needle.

Intraoperative ultrasound plays a critical role in achieving precise surgery [26]. For margin assessment in breast-conserving surgery, computer-assisted ultrasound boundary recognition technology has demonstrated a sensitivity of 96% and specificity of 76% in predicting pathological positivity or margin distance ≤ 2.0 mm, showing promising potential for intraoperative margin evaluation [27]. In the field of real-time tumor segmentation, U-Net-based deep learning models have achieved an average Dice coefficient ranging from 0.78 to 0.959, with an area under the receiver operating characteristic curve (AUC) of 0.94; moreover, 93% of expert visual scores rated ≥ 7/10. The processing speed of 16 frames per second fully satisfies clinical real-time requirements [20, 28]. Regarding biopsy needle guidance, a novel spatiotemporal memory network achieved an Intersection-over-Union (IoU) of 0.731, Dice coefficient of 0.817, and precision of 0.863, significantly enhancing puncture stability and localization accuracy [21]. In the present study, Model 1 achieved a mAP@50 of 0.789 on the test set, while Model 2 reached a mAP@50 of 0.797, indicating high accuracy in target recognition tasks and

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Table 3 Three-fold cross-validation results of Model 1

	Precision (tumor)	Recall (tumor)	map50 (tumor)	Precision (cutter slot)	Recall (cutter slot)	map50 (cutter slot)	Precision (all)	Recall (all)	map50 (all)
1	0.827	0.811	0.864	0.78	0.665	0.716	0.803	0.738	0.79
2	0.831	0.776	0.826	0.825	0.736	0.758	0.828	0.756	0.792
3	0.802	0.686	0.78	0.821	0.765	0.793	0.811	0.725	0.786
Avg	0.82	0.758	0.823	0.809	0.722	0.756	0.814	0.74	0.789

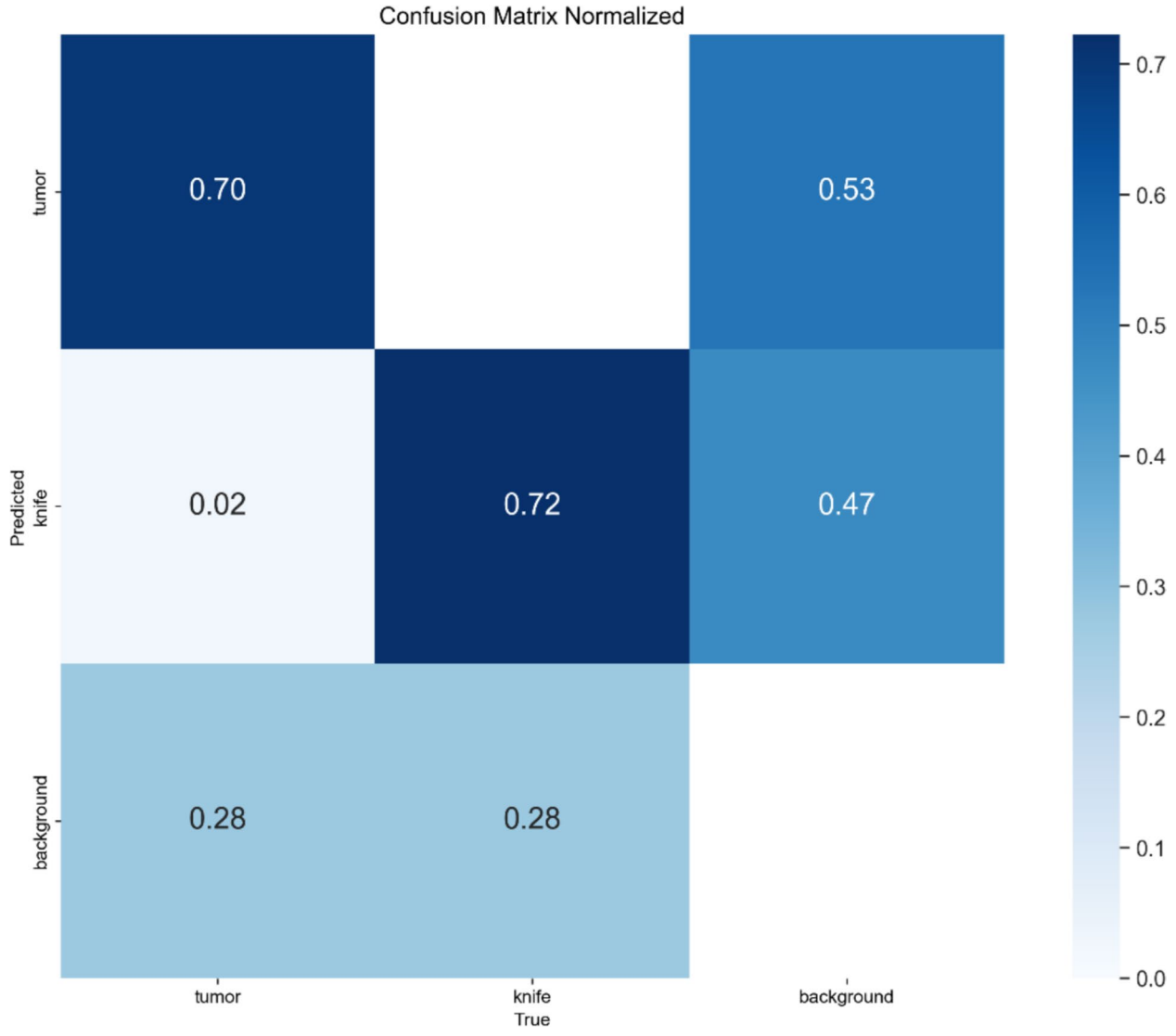


Fig. 2 Confusion matrix of Model 1

effective localization of key intraoperative structures. Such performance meets the basic requirements for clinical application, particularly in scenarios such as breast vacuum-assisted biopsy navigation, which demand both real-time processing and high precision.

VABB is performed entirely under ultrasound guidance, and insufficient ultrasound experience can significantly hinder the completion of the procedure. Halverson et al. reported that 52% of community surgeons expressed little confidence in performing breast ultrasound examinations [9]. Nassour et al. found that only 11.3% of surgical residents had received formal ultrasound training [10]. Mammotome Inc. was the first to design the breast vacuum-assisted biopsy device with two distinct stages: localization and resection. This design facilitates adoption by less experienced junior surgeon [29]. The present study adopted a similar approach by developing Model 1 and Model 2 accordingly. In Model 1, the model's

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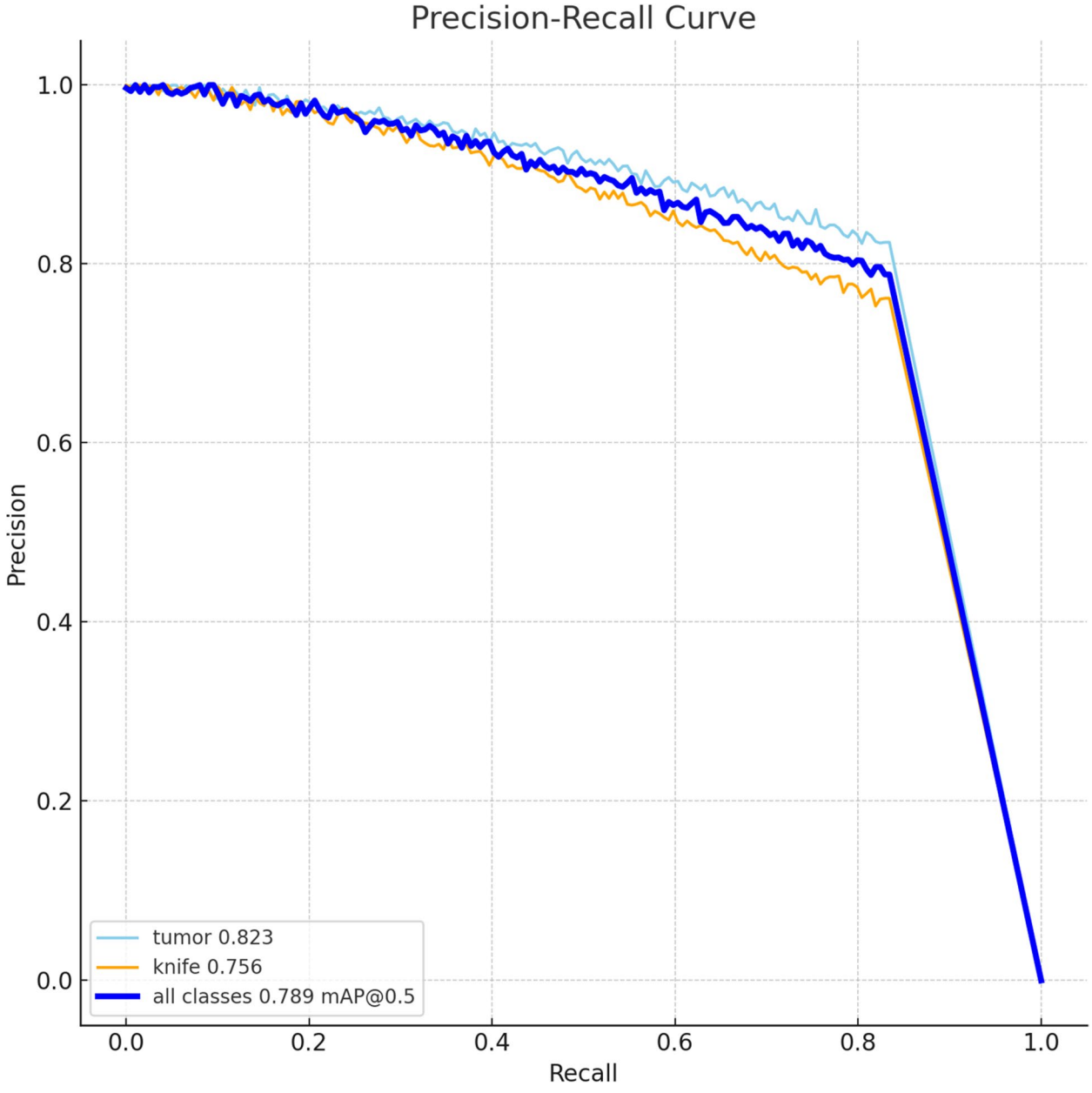


Fig. 3 P-R curve of Model 1

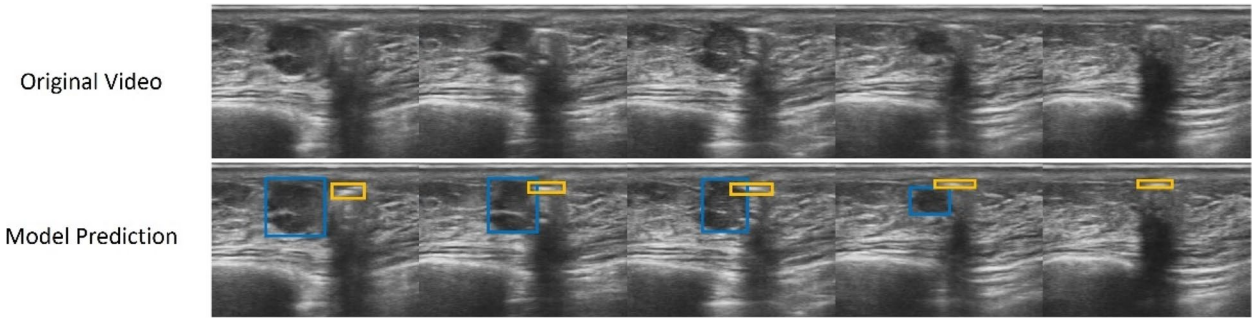


Fig. 4 Schematic diagram of model 2 Recognition (within the same case video stream). Yellow boxes indicate the knife identified by the model in ultrasound videos, while blue boxes denote tumors detected by the model

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Table 4 Performance of Model 2 and comparison with junior surgeon

	Precision(tumor)	Recall(tumor)	map50(tumor)	Precision(knife)	Recall(knife)	map50(knife)	Precision(all)	Recall(all)	map50(all)	AccCoef(all)
Yolo11	0.804	0.753	0.829	0.83	0.749	0.765	0.817	0.751	0.797	0.767
Junior Surgeon	0.785	0.865	0.758	0.502	0.606	0.35	0.644	0.735	0.554	0.599

recognition performance for both tumors and cutter slots surpassed that of junior surgeon, suggesting its potential as an effective assistant during ultrasound-guided breast vacuum-assisted biopsy. Further analysis revealed that both the model and surgeon demonstrated better recognition of tumors than cutter slots, indicating that visualization and identification of the cutter slot remains a technical challenge. First, during the localization phase, the long axis of the rotating cutter is often positioned nearly parallel to the ultrasound probe. Its metallic composition generates substantial acoustic shadowing, interfering with the imaging of adjacent structures [30]. Second, the cutter slot itself lacks distinct ultrasound features and typically requires identification based on "double-line" signs, concavities at the junction between the slot and the cutter body, and estimation of distances between cutter slots. This process heavily relies on the operator's spatial imagination and anatomical recognition skills, posing considerable difficulty for junior surgeon. Moreover, the accuracy of cutter slot identification is constrained by the relative angle and positional control between the probe and the rotating cutter. The narrow probe and slim cutter body mean that slight angular deviations can lead to loss of key imaging features. In Model 2, the model outperformed junior surgeon in recognizing both tumors and cutter slots, with minimal difference in performance between the two targets. This may be attributed to Model 2 detecting cross-sectional images of the cutter, where acoustic shadow interference is reduced and the cutter body is more clearly visualized.

Compared to junior surgeon, the proposed AI navigation system in this study demonstrates significant advantages in the identification and localization of lesions and instruments. The high F1-scores of Model 1 and Model 2 (0.765 and 0.791, respectively) indicate that the system offers high reliability in detecting lesions and instruments, effectively avoiding missed and false detections. Meanwhile, their DSC coefficients (0.714 and 0.751, respectively) have reached a level sufficient for outlining approximate contours during intraoperative navigation. However, we also note that the current models' Hausdorff Distance (HD) remains relatively high, reflecting insufficient boundary segmentation accuracy in some challenging scenarios, which may lead to significant local contour deviations. We posit that the primary role of the model is to provide auxiliary spatial reference for junior surgeons. For this purpose of intraoperative guidance, the current boundary precision, as indicated by the HD metric, is deemed adequate. Nevertheless, further refinement of segmentation accuracy remains a defined objective for our subsequent work.

Therefore, enhancing ultrasound discrimination skills is crucial for performing ultrasound-guided surgical procedures. Regardless of their prior medical background,

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Table 5 Three-fold cross-validation results of Model 2

	Precision(tumor)	Recall(tumor)	map50(tumor)	Precision(knife)	Recall(knife)	map50(knife)	Precision(all)	Recall(all)	map50 (all)
1	0.842	0.859	0.889	0.813	0.731	0.751	0.827	0.795	0.82
2	0.824	0.831	0.885	0.749	0.635	0.668	0.787	0.733	0.777
3	0.832	0.839	0.873	0.811	0.721	0.743	0.821	0.78	0.808
Avg	0.832	0.843	0.882	0.791	0.695	0.72	0.811	0.769	0.801

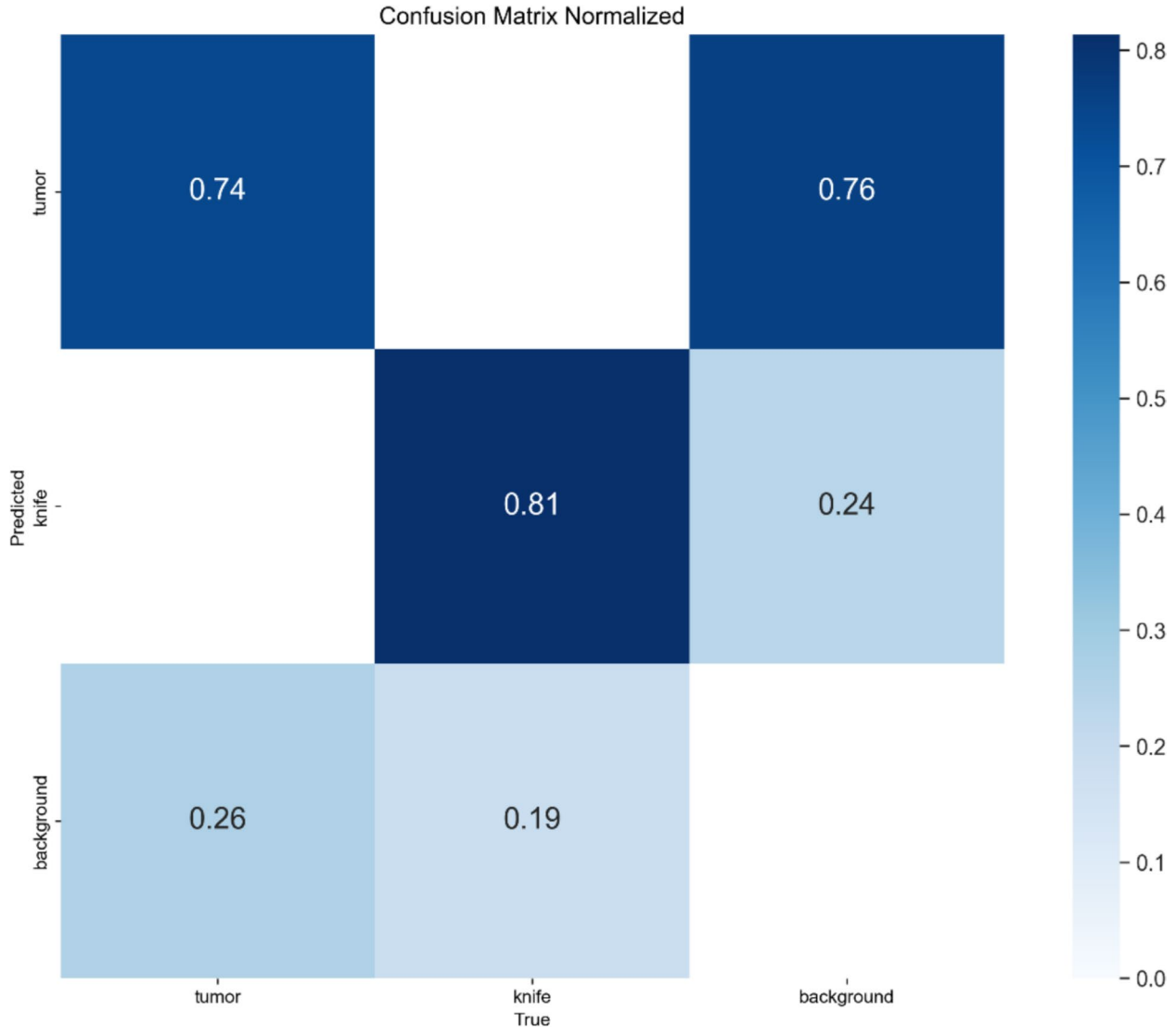


Fig. 5 Confusion matrix of Model 2

even with just two weeks of radiology rotation training, students can significantly improve their ultrasound skills [31]. Moreover, practical exercises with the model can further elevate the surgical proficiency of trainees. A systematic evaluation of practical training for ultrasound-guided minimally invasive breast biopsy demonstrated that the use of simulation models in training significantly outperformed traditional lecture-based teaching in surgical skill scores and localization time [32]. Finally, the surgical navigation model developed in this study also holds promise for further expansion as an educational tool for preoperative simulation and skill training.

The model used in this study contains 9,413,574 parameters, maintaining strong performance while ensuring lightweight design suitable for deployment in resource-constrained environments. With a computational complexity of 21.3 GFLOPS, the model exhibits low computational demand, facilitating real-time processing.

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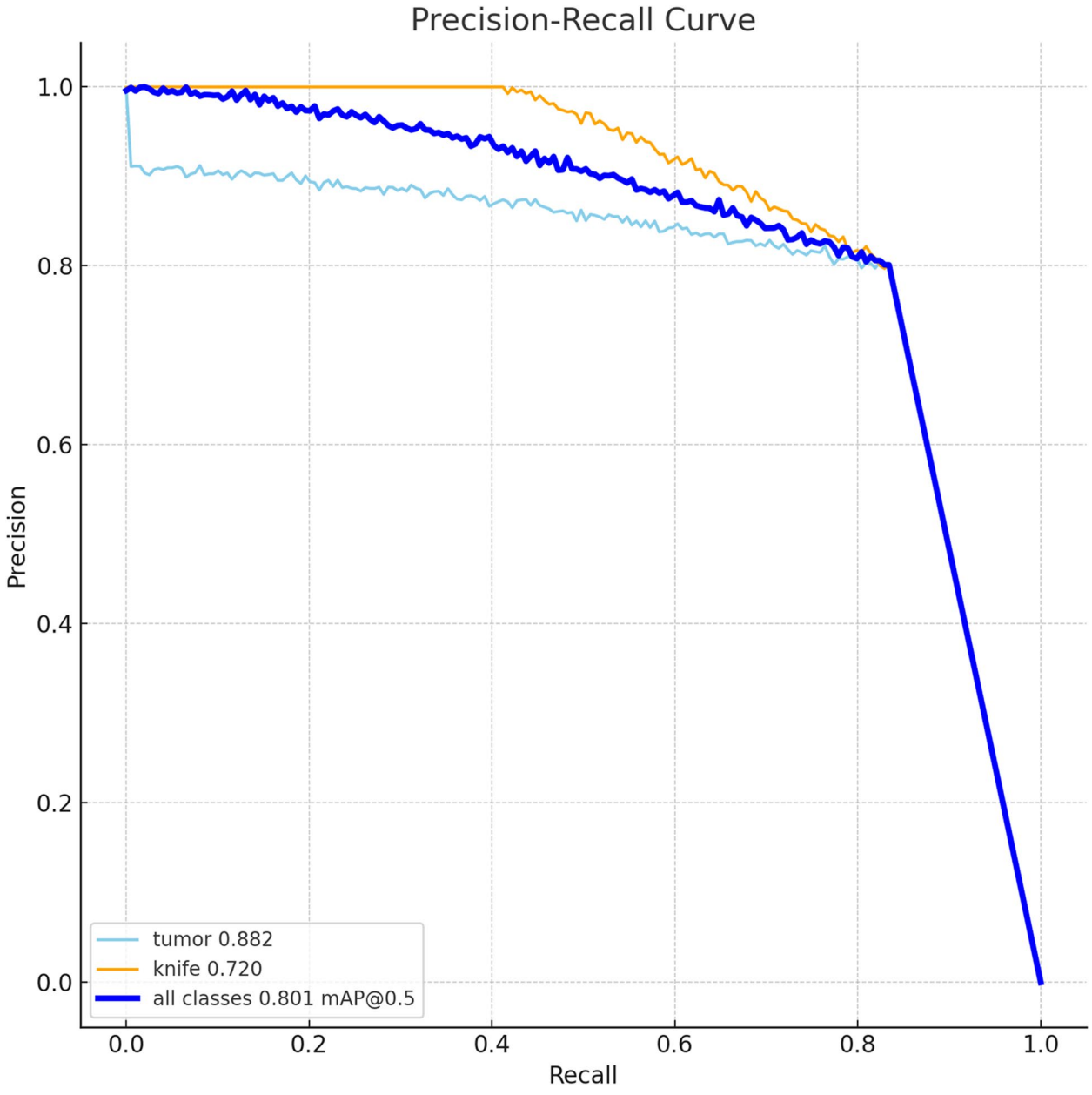


Fig. 6 P-R curve of Model 2

On the GPU platform, inference time is only 1.2 ms per frame, demonstrating its potential for high-timeliness applications such as intraoperative navigation. Inference on the CPU platform requires approximately 32.6 ms per frame, meeting the real-time requirements of breast vacuum-assisted biopsy navigation. Future improvements in system responsiveness may be achieved through model compression, parallel optimization, or heterogeneous acceleration techniques (e.g., combining GPU and NPU).

This study has the following limitations: (1) The cases used for model training were sourced from a single institution. Although a sufficiently large dataset was provided for model training, future studies will implement secure multi-center medical data sharing through federated hospital networks to enhance the generalizability and robustness of the AI model while ensuring patient privacy protection. (2) This study aimed to assess the feasibility of the model. However, further research is needed to evaluate its safety and effectiveness in breast vacuum-assisted biopsy procedures. We are planning a new study to investigate whether the use of our AI navigation model during surgery can reduce blood loss and other complications. Additionally, we will test the model's real-time latency, frame rate, and gather feedback from surgeons to assess its practicality in a clinical setting. (3) Although YOLOv11 achieved promising recognition accuracy in

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breast vacuum-assisted biopsy navigation, there remains room for improvement. Future research will focus on technical optimization and the development of a lightweight model architecture.

In practice, 3D ultrasound provides more detailed three-dimensional images, which are particularly useful for locating complex anatomical structures, demonstrating significant advantages in clinical settings [33]. Robotic-assisted ultrasound, through precise control of the probe position, can greatly reduce errors caused by operator fatigue or instability, thereby providing more stable real-time image guidance [34]. Furthermore, existing robotic surgical platforms, such as the da Vinci system, can be integrated with ultrasound image guidance systems to achieve more efficient and accurate surgical procedures [35]. Given the superiority of these technologies, future work could explore how to combine these existing devices with the model we have proposed.

Conclusion

The deep learning model proposed in this study demonstrated high accuracy in recognizing the knife tip and tumor during the VABB, outperforming junior surgeon. The YOLOv11 network achieved detection accuracies of 0.829 mAP50 for tumors and 0.765 mAP50 for the knife tip on a dataset of 10,668 ultrasound images, exceeding the junior surgeon' performance (0.758 and 0.350, respectively). Additionally, the model exhibited efficient computational performance, with measured inference times of 32.6 ms on the CPU platform and only 1.2 ms on the GPU platform, meeting the standards for deployment on portable devices. These results indicate that our AI model holds strong potential for application in real-time navigation systems for breast vacuum-assisted biopsy procedures.

Abbreviations

VABB	Vacuum-assisted breast biopsy
AI	Artificial intelligence
mAP50	Mean Average Precision at 0.5 Intersection-over-Union
IoU	Intersection-over-Union
PR curve	Precision-recall curve
IQR	Interquartile range
HD	Hausdorff Distance
DSC	Dice Similarity Coefficient

Acknowledgements

Luo Haibo, Shenyang Institute of Automation, Chinese Academy of Sciences
Hailong Wang, Department of Clinical Epidemiology, China Medical University

Authors' contributions

Conceptualization, Xinran Shao; Data curation, Ting Ruan, Xiang Fei and Jianchun Cui; Formal analysis, Yunzhi Shen], Ting Ruan]; Funding acquisition, Xingai Ju; Investigation, Jinrui Liu and Xiang Fei; Methodology, Pingdong Sun] and Qiushi Li; Project administration, Xingai Ju; Resources,]Xiang Fei and Jianchun Cui; Software, Qikun Guo、Yuxin Ma; Supervision, Xingai Ju; Validation, Yihan Sun and Jinrui Liu; Visualization, Yuqing Wangand Jianchun Cui; Writing – original draft, Pingdong Sun, Xinran Shao, Yihan Sun and Xiang Fei; Writing – review & editing, Yunzhi Shen, Qiushi Li, Ting Ruan, Jinrui Liu, Xiang Fei, Xingai Ju and Jianchun Cui. Xinran Shao、Yunzhi Shen, Pingdong Sun、Yihan Sun、Xingai Ju contributed equally to this work.

Funding

This work was supported by The Doctoral Scientific Research Startup Foundation of Liaoning Provincial Joint Fund Program (Grant No. 2023-BSBA-181). The Applied Basic Research Program of Liaoning Provincial Department of Science and Technology (Grant No. 2022020247-JH2/1013).

Data availability

The dataset supporting the conclusions of this article is included within the article (and its additional file).

Declarations

Ethics approval and consent to participate

The study protocol was reviewed and approved by the People's Hospital of China Medical University ethics committee No. 2023H005. Informed consent was obtained from all patients enrolled in the study.

Competing interests

The authors declare no competing interests.

Author details

[1]Department of Thyroid and Breast Surgery, People's Hospital of China Medical University（People's Hospital of Liaoning Pronvince), Shenyang, China
[2]Dalian Medical University, Dalian, China
[3]Department of Cardiology, People's Hospital of China Medical University（People's Hospital of Liaoning Pronvince), Shenyang, China
[4]Department of General Medicine, People's Hospital of China Medical University（People's Hospital of Liaoning Pronvince), Shenyang, China
[5]The College of Medicine and Biological Information Engineering, Northeastern University, Shenyang, China
[6]Liaoning University of Traditional Chinese Medicine, Shenyang, China
[7]The Graduate School of Artificial Intelligence, China Medical University, Shenyang, China
[8]The School of Information Science and Engineering, Shenyang Ligong University, Shenyang, China

Received: 21 June 2025 / Accepted: 17 October 2025
Published online: 13 November 2025

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